ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba [ANSWER KEY]

Unit 4.01: Bijective Counting, Stars & Bars, PIE & Pigeonhole

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 12x + 50 in the domain of Bijective Counting, Inclusion-Exclusion & Pigeonhole. By completing the square f_1x = x - h^2 + k , find the sum k + f_18 .

[ ] A: 38
[ ] B: 18
[ ] C: 30
[x] D: 32

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Complete the square: x^2 - 12x + 50 = x - 6^2 - 36 + 50 = x - 6^2 + 14 , giving minimum value k = 14 . **[M1]** 2. Evaluate at x = 8 : f_18 = 2^2 + 14 = 18 . **[M1]** 3. Sum the two quantities: k + f_18 = 14 + 18 = 32 . **[A1]** - **Final Verified Answer:** 32 .

Q2. A mathematical model in Bijective Counting, Inclusion-Exclusion & Pigeonhole is defined by the cubic function y_2x = 2x^3 - 6x^2 + 15x - 11 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[x] A: 33
[ ] B: 69
[ ] C: 18
[ ] D: 21

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 6x^2 - 12x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 69 - 123 + 15 = 33 . **[M1][A1]** - **Final Verified Answer:** 33 .

Q3. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, two displacement vectors are given by mathbf{u}_{3} = langle 6, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -2
[x] B: 98
[ ] C: 100
[ ] D: 210

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 69 + -87 = 54 + -56 = -2 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 6^2 + -8^2 = 100 . **[M1]** 3. Combine: -2 + 100 = 98 . **[A1]** - **Final Verified Answer:** 98 .

Q4. The linear growth profile shown in the diagram for Bijective Counting, Inclusion-Exclusion & Pigeonhole follows an arithmetic progression with first term a_1 = 7 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 91
[ ] B: 825
[x] C: 735
[ ] D: 1365

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 7, d = 6 : S_{15} = frac{15}{2}[14 + 14 times 6] = frac{15}{2} times 98 = 735 . **[M1][A1]** - **Final Verified Answer:** 735 .

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Q5. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, evaluate the exact definite integral $I_5 = \int_0^{2} 18x^2 + 24x\,\mathrm{d}x$.

[ ] A: 120
[ ] B: 120 + 1
[ ] C: 49
[x] D: 96

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Integrate term by term: $\int 18x^2 + 24x\,\mathrm{d}x = \left[6x^3 + 12x^2\right]_0^{2}$. **[M1]** 2. Substitute the upper and lower limits: $62^3 + 122^2 - 0 = 48 + 48 = 96$. **[M1][A1]** - **Final Verified Answer:** 96 .

Q6. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, consider the quadratic function $f_{6x} = x^2 - 14x + 135$. Express f_{6x} in the completed-square form $x - h^2 + k$ and compute the exact value of $k + f_{614}$.

[x] A: 221
[ ] B: 228
[ ] C: 135
[ ] D: 214

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Complete the square: $x^2 - 14x + 135 = x - 7^2 - 49 + 135 = x - 7^2 + 86$, giving minimum value $k = 86$. **[M1]** 2. Evaluate at $x = 14$: $f_{614} = 7^2 + 86 = 135$. **[M1]** 3. Sum the two quantities: $k + f_{614} = 86 + 135 = 221$. **[A1]** - **Final Verified Answer:** 221 .

Q7. The graph illustrates the cubic model $y_{7x} = 2x^3 - 7x^2 + 33x - 18$ studied in Bijective Counting, Inclusion-Exclusion & Pigeonhole. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

[ ] A: 417
[x] B: 305
[ ] C: 272
[ ] D: 67

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 14x + 33$. **[M1]** 2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 664 - 148 + 33 = 305$. **[M1][A1]** - **Final Verified Answer:** 305 .

Q8. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, two displacement vectors are given by $\mathbf{u}_{8} = \langle 7, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + |\mathbf{u}_{8}|^2$.

[ ] A: 21
[ ] B: 98
[x] C: 119
[ ] D: 189

Solution: **Mark Scheme & Examiner Solution Bijective Counting, Inclusion-Exclusion & Pigeonhole:** 1. Compute the inner product: $\mathbf{u}_{8} \cdot \mathbf{v}_{8} = 78 + -75 = 56 + -35 = 21$. **[M1]** 2. Compute the squared norm: $|\mathbf{u}_{8}|^2 = 7^2 + -7^2 = 98$. **[M1]** 3. Combine: $21 + 98 = 119$. **[A1]** - **Final Verified Answer:** 119 .